

Data 102 Final Project Written Report
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Dataset Overview

The dataset used for this paper comes from the provided election dataset, FiveThirtyEight's 2022 Primary Candidate Endorsement, which includes information on candidates running in primary elections in the United States in 2022. The data is a census of major candidates in 2022 primaries, not a small random sample, and it includes almost all candidates from major parties in primaries for House, Senate, and Governor positions elected in 2022. While not excluding any particular groups on a large level, the data is focused on significant primaries and thus doesn't include small, local races, and also may not include candidates who dropped out early or some independent candidates. This brings up the potential of selection bias/convenience sampling; the data only includes races that were considered significant enough to track, so if these races happen to reveal more patterns due to raised competitiveness or news coverage, our analyses may not be applicable to all primaries regardless of size, which is an important consideration. The sample is not randomly generated, so the typical notions about random samples can't be applied blindly here. However, the size and comprehensiveness of the data makes it a completely reasonable dataset to analyze as long as our conclusions acknowledge the possible limitations of the data. Ideally, the dataset would also include some information about specific cost breakdown of campaign spending, voter demographics, and other details that would lead to more confidence in our analyses, but the depth of the data still allows for causal analysis and considerable comparisons for hypothesis testing.

For our analysis, we primarily used Republican and Democratic candidate files, where each row represents a single candidate in a specific primary election. Each row contains information such as primary vote share, whether the candidate won or lost, gender, race, incumbency, office, state, and for Republicans, whether or not they were endorsed by Donald Trump. The granularity being candidates and not voters led us to focus most of our analysis on candidate success and patterns leading to higher vote share. The data did include some missing values, so for any test involving a variable that was missing from a row, or too ambiguous to work with, the row was discounted to avoid misleading results. Cleaning the data involved removing rows with missing values for specific tests, removing whitespace and other characters to make certain columns numeric, binarizing certain columns, and other general work to make the data more workable for our purposes.

Research Questions

Our project uses two methods of data analysis, causal inference and multiple hypothesis testing, to answer two research questions about candidate win patterns. The first research question focuses on causal effect: Among Republican candidates in the 2022 primary elections, what is the causal effect of receiving an endorsement from Donald Trump on a candidate's chance of winning compared to candidates without an endorsement? Analyzing data to answer this question can inform real-world decisions, as understanding the causal effect of an endorsement can impact Republican candidates and their campaign teams. The results can inform them of how impactful an endorsement is, which can influence them to either pursue an endorsement or choose not to, based on their current election standings

and their goals. Donors and voters may also be swayed by the results, as seeing the causal impact of an endorsement may make them reconsider a factor they may have otherwise ignored. Causal inference is a good fit for this question because we are explicitly looking for the effect of a binary variable (whether or not a candidate has the endorsement). This binary variable can be considered a treatment, so by testing for causality, we are essentially testing the effect of the treatment of a Trump endorsement on a binary outcome (win versus loss). There are several factors such as incumbency and state that can confound the effect of the endorsement on winning, but these are taken into account by balancing treated and untreated candidates using inverse probability weighting. Apart from confounders, there are still some limitations to using causal inference in this case; endorsements are not randomly assigned, so our analysis assumes that there aren't any significant, unmeasured confounders affecting the chance of receiving an endorsement or the chance of winning the election. Factors such as media coverage, personal/business connections, and other possible confounders are not considered in the data, so using causal inference here comes at the cost of making certain assumptions about the elections. These limitations and their implications are worth recognizing and discussing, but they do not discredit the analysis as a whole, and causal inference is still the most applicable method to apply for this question.

The second research question focuses on multiple hypothesis testing and comparisons: In 2022 primary elections, are there significant differences in primary win rates across gender, race, and incumbency status across parties, and which differences are statistically significant after multiple comparisons? Answering this question addresses real world questions about what demographic information does or doesn't impact election outcomes, which is valuable for candidates to know before they choose to run. It's also valuable to know societally, as certain demographics being systematically voted for less is a concerning statistic that warrants further examination. Multiple hypothesis testing is the best fitting method here, because we are explicitly running several different, yet related, tests, including gender differences by party, race differences, and more. For each hypothesis, we run a two-sample proportion test comparing win rates between the two groups being tested, and then apply the Benjamini–Hochberg correction to control the false discovery rate. Though beneficial overall, multiple hypothesis testing and correction methods do come with some limitations; Benjamini-Hochberg allows for more discoveries overall, but it also allows for a controlled proportion of false positives. The comparisons made in hypothesis tests, though significant and worth analyzing, are not necessarily causal, and are descriptive, observed differences, not definitive causal effects. This distinction is important to keep in mind when discussing the results of the hypothesis tests.

Prior Work

Our first research question about the causal effect of a Trump endorsement on winning is related to several questions that have been explored in prior works. The discussion of causal effects of endorsements is well documented, and works such as Boudreau's 2016 paper, "The Persuasion Effects of Political Endorsements" study exactly that. Boudreau explores how endorsements impact voters and their decision making, as well as election outcomes. This paper is relevant to our question because it addresses the same overarching question about the effects of endorsements on winning, just in a broader sense than ours, which is focused on specifically Trump endorsements. Another paper, Blatte's 2024 "The Causal Effects of a Trump Endorsement on Voter Preferences in a General Election Scenario" is even more related, as it asks essentially the same exact question, just in a slightly more general sense. Blatte

discovers that Trump's endorsement can meaningfully impact voter preferences towards endorsed candidates, usually through experimental design and surveys focused on general elections. Our specificity and methodology differs from Blatte, as we are focused solely on primary elections, and we are using observational data to draw conclusions rather than performing surveys or experiments. Our work differs methodologically from both papers, revealing some limitations in our analysis, which is not as in depth as the experimental analysis of the two papers, but it also emphasizes how our work is different from other prior works, showing the relevance and importance of our findings.

The second research question, involving hypothesis testing, also has elements that can be seen in prior works. The concept of demographics such as gender and race impacting election outcomes has been highly discussed and analyzed over time, especially as more and more female and POC candidates have appeared in recent times. Anzia and Bernhard's 2022 paper, "Gender Stereotyping and the Electoral Success of Women Candidates: New Evidence from Local Elections in the United States," analyzes the results of thousands of local elections, testing if women's and men's win rates match theoretical ideas and expectations about gender stereotyping. Several of our own hypotheses are centered around gender and its influence on election outcomes, although our data is from larger, primary elections from 2022 rather than small local elections. Overall, we're asking very similar questions, essentially seeing if the patterns from their findings on local elections extend into our findings on larger primaries. Their dataset is significantly larger and more encompassing, which does show a shortcoming in our analysis, but the difference in type of election highlights the unique angle our research is taking. A similar article from Stanford News, titled "Groupthink Gone Wrong: Assumptions About Electability Undermine Women Political Candidates" (2022), studies how perceptions that women are less electable can have a huge impact on voters and donors, leading to female candidates receiving much less support despite their qualifications. Once again, this article is studying the same pattern as many of our hypotheses: how does gender play a role in election outcomes? Though similar thematically, our research takes a different approach than these prior works, as both of these papers focus more on voter perception and behavior, while our analysis focuses solely on election outcome.

EDA

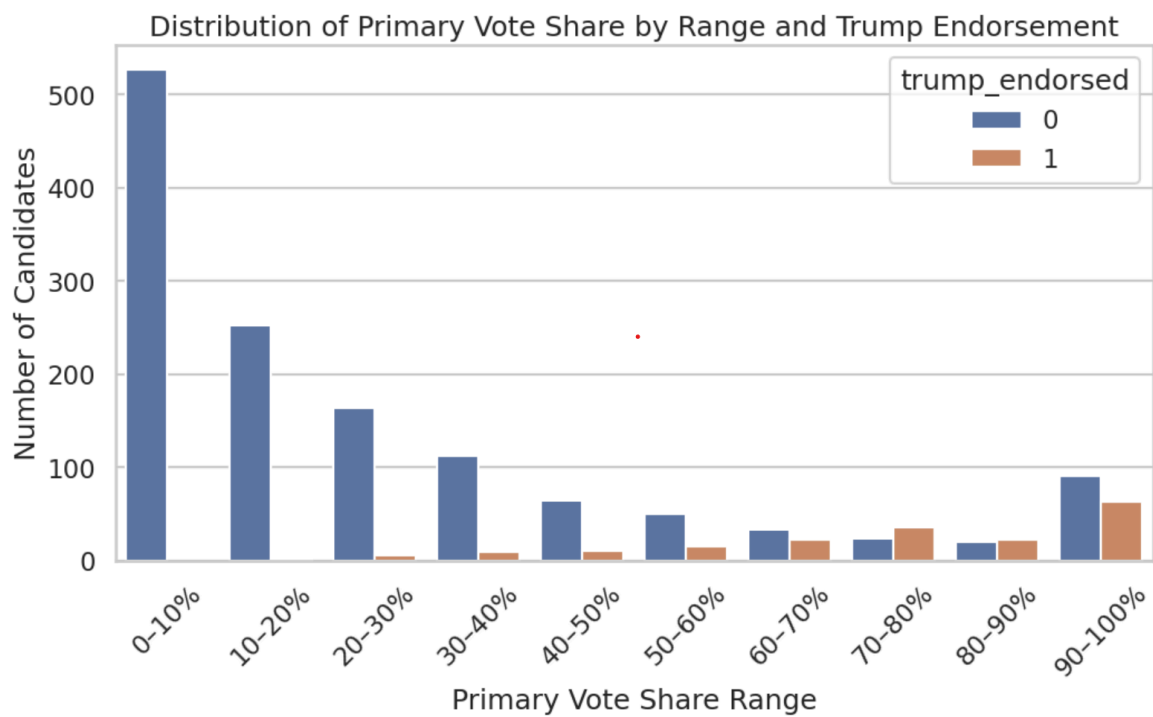
The exploratory data analysis process is used to explore distributions and potential confounders within the data set. Ultimately, we seek to understand descriptively the differences, similarities, and trends seen within groups to motivate our methodology for the research questions at hand.

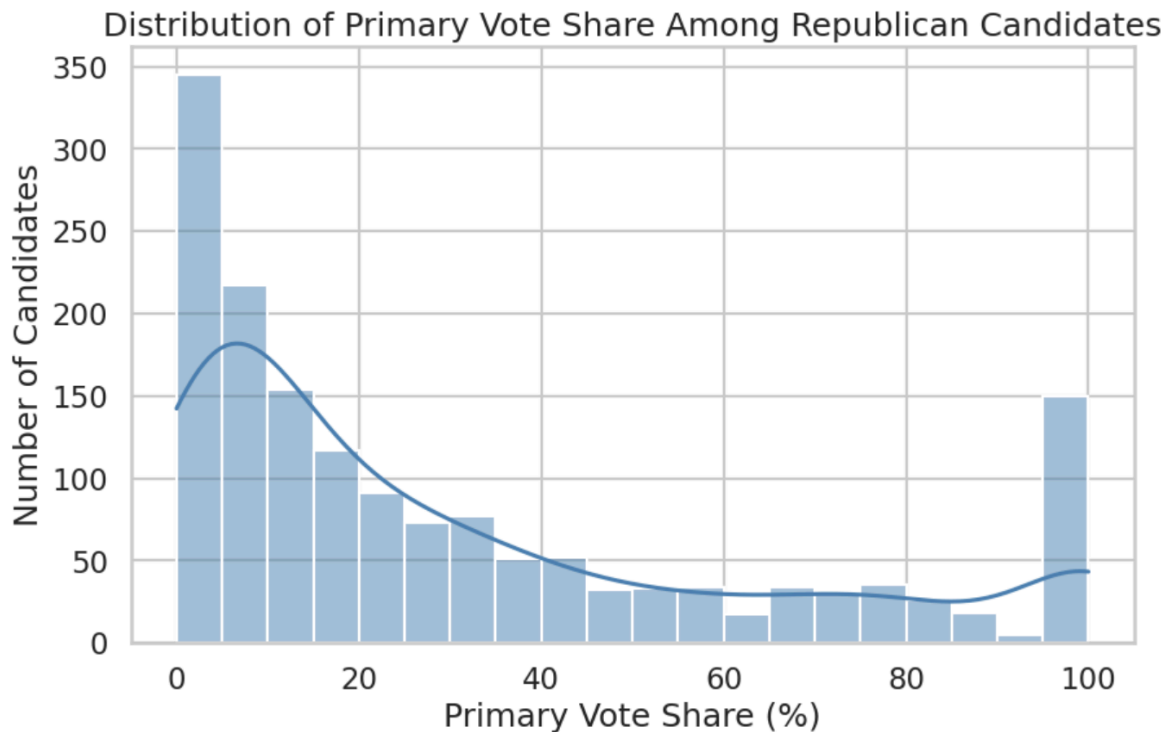
EDA Question 1

••

count	
trump_endorsed	
0	1409
1	190

dtype: int64

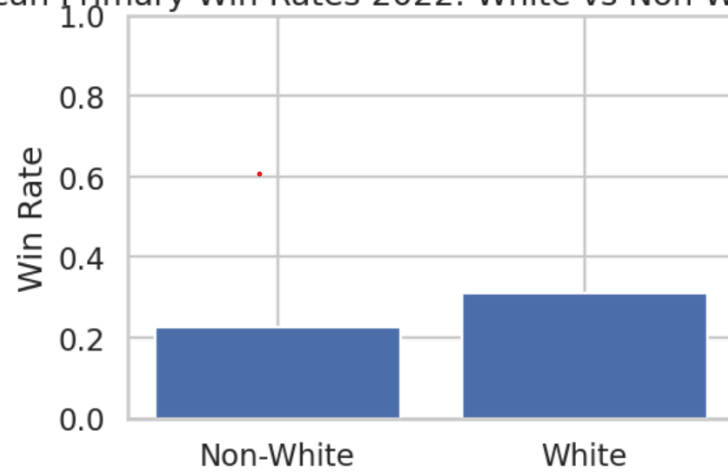




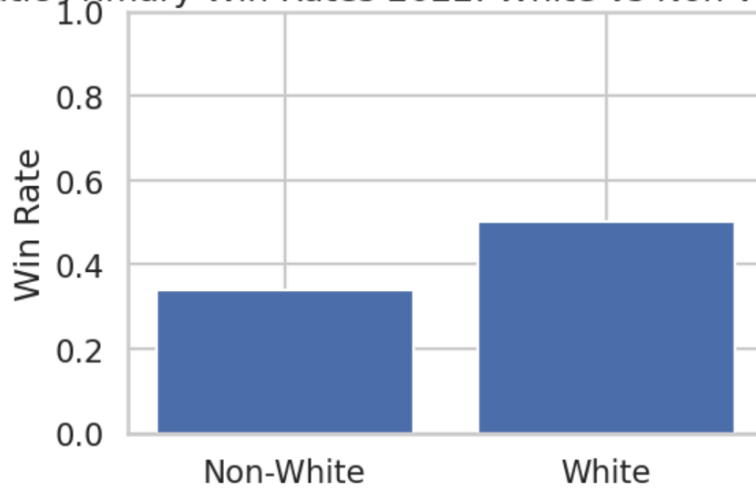
In our EDA analysis of the first research question, which asks if there is a causal effect of receiving a Trump endorsement on a candidate's chance of winning in the 2022 primary election?, our population is restricted to the Republican candidates, who are the prospects that can or can not receive a Trump endorsement. Further looking at the distribution of endorsements amongst Republicans, we see it seems relatively uncommon to receive one, with nearly 13.4% (190/1490) receiving endorsement, revealing an imbalance for the number of endorsements to no endorsements. We then crafted an overlaid histogram to get the number of candidates (endorsed vs not endorsed) under receiving a range of primary vote shares. The plot reveals a skew. Not endorsed candidate distribution is more on the left and thus clustered around the lower primary vote share ranges. Endorsed candidates distribution is more on the right end and hence in the higher primary vote share ranges. Although this EDA process gives us a sense of strong association between endorsement on winning chances, confounding in endorsements from systemic differences (geographically, race, gender), reinforce the need for causal inferences in understanding the research question.

EDA Question 2

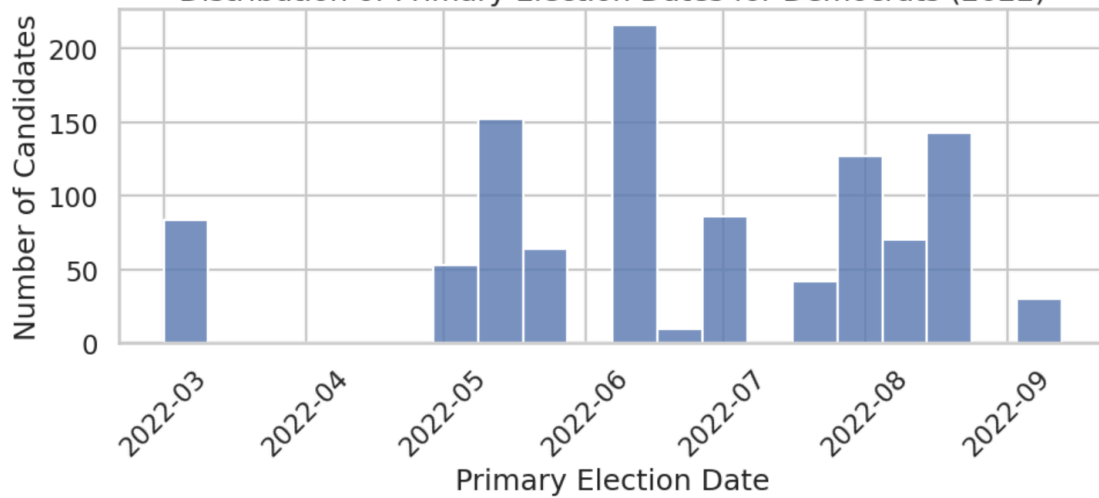
Republican Primary Win Rates 2022: White vs Non-White Candidates

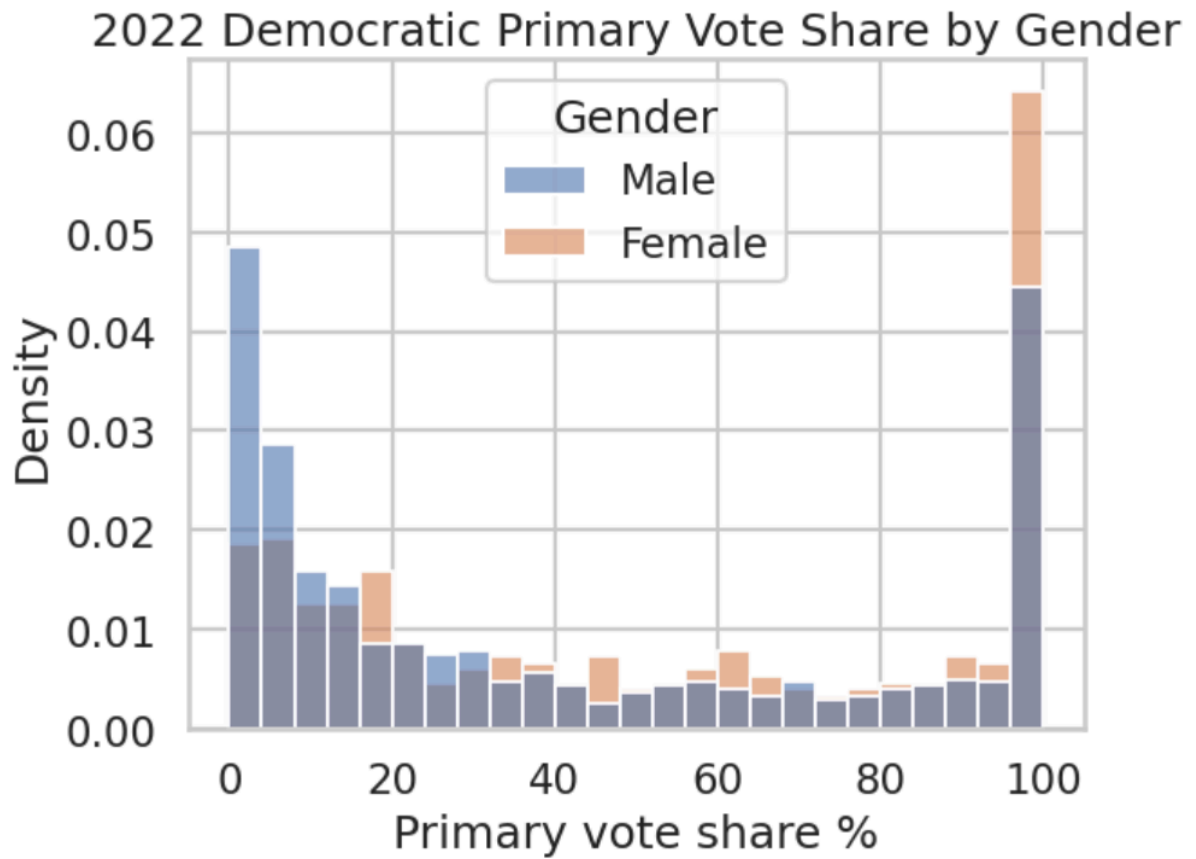
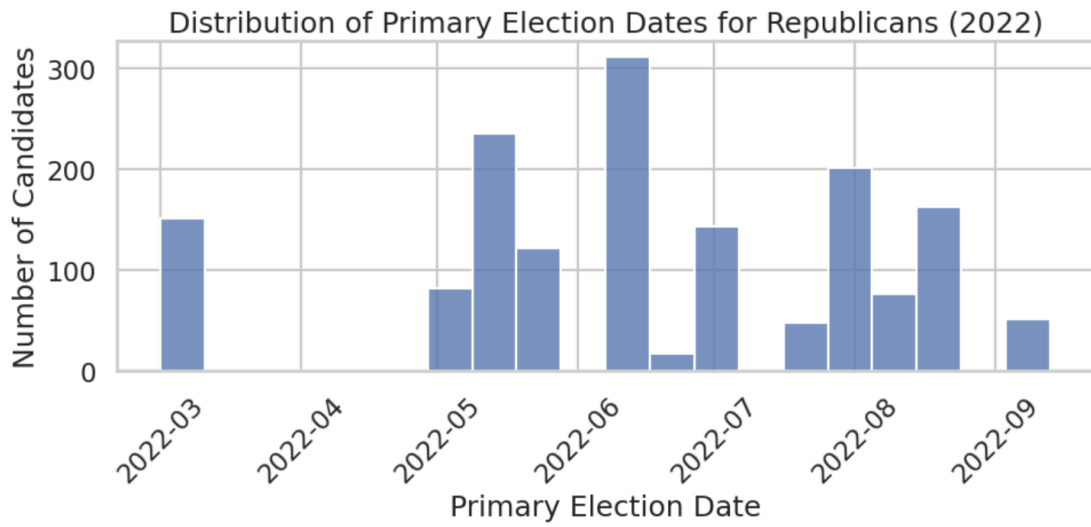


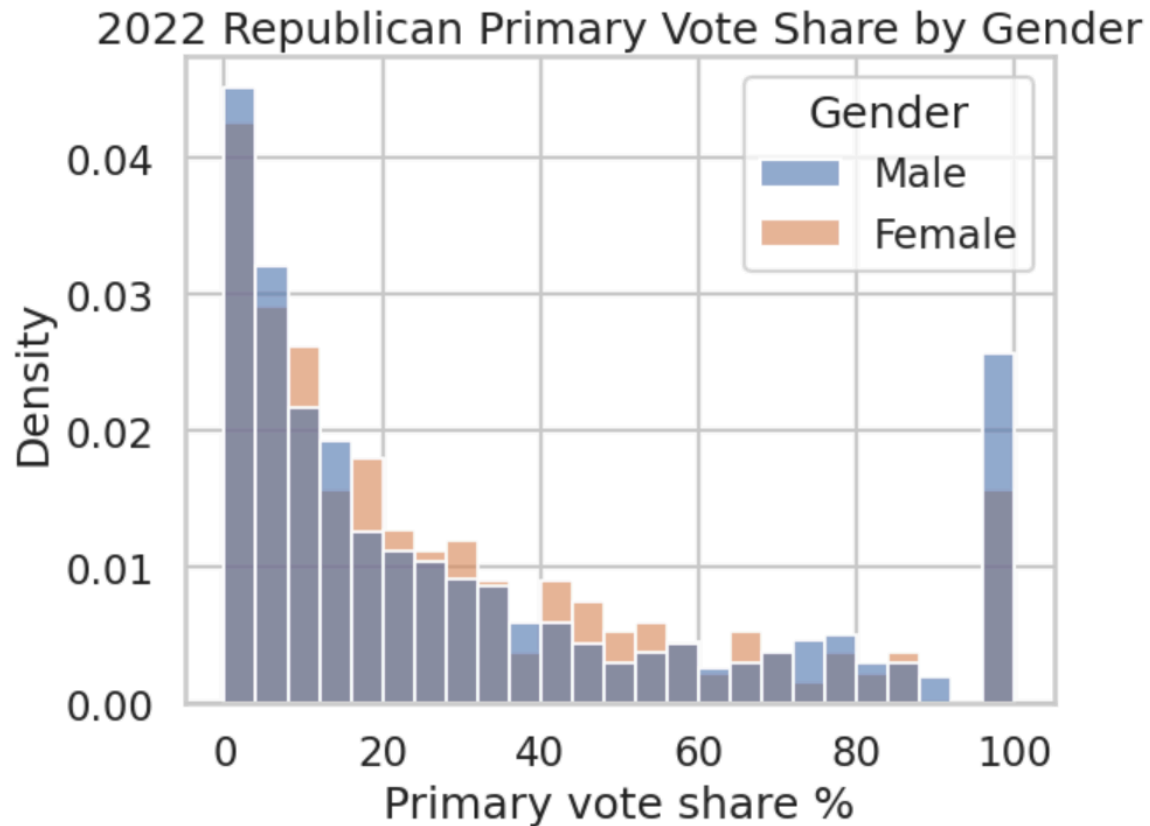
Democratic Primary Win Rates 2022: White vs Non-White Candidates



Distribution of Primary Election Dates for Democrats (2022)







In our EDA analysis of the second research question, which asks if there are any visible differences in primary win rates across various candidates in demographical/positional groups (including gender, race, current candidate position, etc), we approached with the intent of using multiple hypothesis testing for different scenarios. We began with seeing what the data shows about differences in win rates by race separated by political party. We notice the most difference in win-rate for white vs non-white candidates in the democratic party, which suggests that patterns can vary depending on the party in question and bring group-based comparisons into interest. We also visualized separate distributions for the democratic and republican party of primary election dates to provide a temporal structure of our data. Both plots revealed that primary elections are more concentrated at certain points in time for both parties. Although this does not directly lead to our hypothesis, it helps contextualize the dataset and helps ensure that comparisons across the political parties are less influenced by time. The last part of our EDA analysis was primary vote share by gender for each party separately which revealed some descriptive differences. For the democratic party, female candidates had higher overall distributions of note share percentage than male, but the opposite was true for the Republican party. This visualization allows us to consider the impact of gender on elections for both parties. After finding multiple group differences through this EDA process, we were able to hypothesize plausible comparisons for the multiple hypothesis process.

Research Question 1: Causal Inference

Our first research question via causal inference is as follows: In the 2022 primary elections, is there a causal effect of receiving an endorsement from Donald Trump on a candidate's chance of winning among other Republican candidates?

Methods:

For this research question, our overarching framework is to define treatment, receiving a Trump endorsement, as a binary indicator. Hence, $T = 1$ occurs if the candidate received a Trump endorsement and 0 otherwise. The outcome, primary election outcome, is also treated as binary, with 1 referencing "won" and 0 referencing "lost."

Our identified confounders are inclusive of the following:

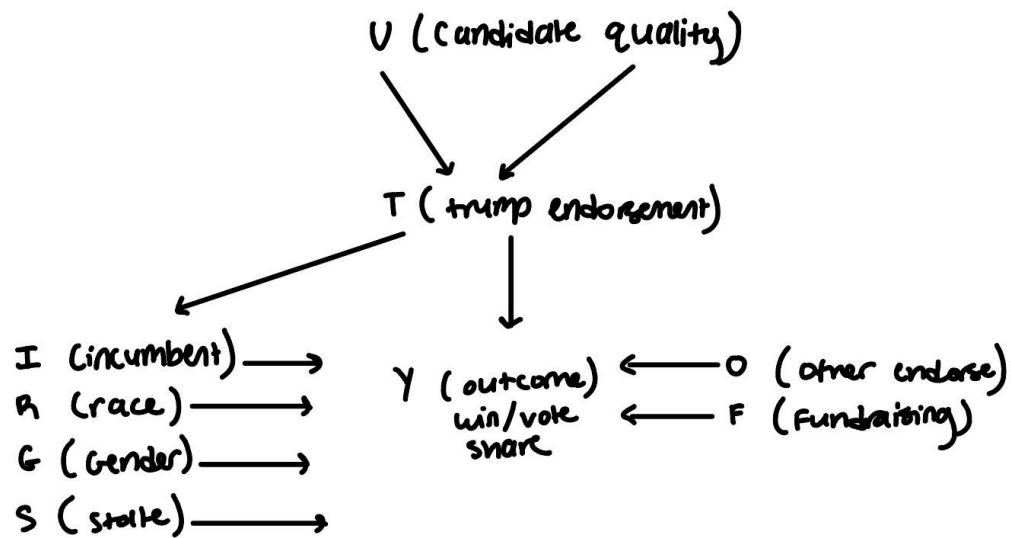
- I: Incumbency
- R: Race
- G: Gender
- S: State/Region
- **F: Fundraising/PAC money**
- **O: Other Endorsements**
- U: any Unobserved candidate qualities
- I, F, & O → these candidates are more attractive for Trump to endorse and are more likely to win even without his support
- S → state and region capture differences in baseline support for Trump and in primary elections.
- R & G → may shape both endorsement strategies & voter behavior

We assume unconfoundedness: conditional on these observed confounders, Trump endorsement is as-if random. Under this assumption, adjusting for these variables using propensity scores and Inverse Propensity Weighting (IPW) lets us interpret differences in win rates as the causal effect of Trump endorsement. Although we adjust for several observed confounders, unobserved confounders may violate the assumption.

Our identified colliders are inclusive of the following:

- Final vote total
- Primary Outcome & Runoff Votes D
- Post-endorsement funding
- Trump Date: timing of endorsement

Our DAG (directed acyclical graph) reflects how these components interact:



Using the Republican candidate data, we estimated the causal effect of Trump endorsement on the probability of winning the primary among other republican candidates. We first estimated a propensity score for each candidate, $e(X_i) = P(T_i = 1 | X_i)$, based on observed confounders: incumbency, race, gender, state/region, fundraising, and other endorsements. We then trimmed propensity scores to $[0.05, 0.95]$ to avoid extremely large IPW weights, and computed inverse propensity weights: $1/e(X_i)$ for endorsed candidates and $1/(1-e(X_i))$ for non-endorsed candidates.

Results:

```
***
                        Generalized Linear Model Regression Results
=====
Dep. Variable:          won_primary      No. Observations:          1599
Model:                  GLM              Df Residuals:              2080
Model Family:           Binomial         Df Model:                  1
Link Function:          Logit            Scale:                    1.0000
Method:                 IRLS             Log-Likelihood:            -1093.7
Date:                   Wed, 17 Dec 2025 Deviance:                   2187.5
Time:                   03:18:25         Pearson chi2:              2.08e+03
No. Iterations:         4                Pseudo R-squ. (CS):        0.3025
Covariance Type:        nonrobust
=====
                        coef      std err          z      P>|z|      [0.025      0.975]
-----
const                -1.1852      0.060     -19.875      0.000     -1.302     -1.068
trump_endorsed         2.7180      0.130     20.916      0.000         2.463         2.973
=====
```

The IPW ATE is defined as the difference in reweighted mean win outcomes between treated and control candidates. In our data, this estimate is about 0.035, meaning that in the reweighted “as-if randomized” population, Trump endorsement is associated with roughly a 3–4 percentage-point higher average probability of winning the primary. This gives a risk-difference interpretation: on average, endorsement increases win probability by a few percentage points after adjusting for the observed confounders.

To further summarize the relationship, we fit a weighted logistic regression (binomial GLM) of the binary outcome, `won_primary`, on the treatment indicator, `trump_endorsed`, using the IPW weights as frequency weights. The estimated log-odds coefficient on Trump endorsement is about 2.72 (95% CI $\approx [2.46, 2.97]$, $p < 0.001$), so the effect is statistically significant. In the reweighted pseudo-population, this corresponds to an estimated win probability of around 23% for otherwise similar non-endorsed candidates versus about 82% for endorsed candidates, a model-based difference of roughly 60 percentage points (odds ratio ≈ 15). This contrast is on a different scale than the IPW ATE (a model-based probability gap at the fitted intercept rather than an average risk difference over all candidates), so the two numbers are not expected to match exactly, but both the IPW estimator and the logistic regression point to a positive, statistically significant, and substantively large association between Trump endorsement and winning, under the assumptions described in the methods.

Uncertainty & Interpretation:

Uncertainty in our estimates comes from both sampling noise and modeling choices. The very small p-value in the weighted logistic regression suggests that, conditional on our model and IPW weights, we can statistically rule out a zero effect of Trump endorsement on win probability, but the exact magnitude of the effect is sensitive to how we specify and trim the propensity scores and to the choice of outcome model (IPW ATE vs. weighted logistic regression). As a result, the numerical values (such as the 3–4 percentage-point IPW ATE and the larger model-based gap in predicted win probabilities) should be viewed as approximate rather than exact. Our causal interpretation also relies on the unconfoundedness

assumption from class: conditional on incumbency, demographics, region, and fundraising/other endorsements, Trump endorsement is treated as as-if random; if important factors like candidate quality or campaign strength remain unobserved, this assumption may be violated and our estimates could still be biased.

Discussion:

This result is suggestive that Trump's endorsements are consistent with a positive causal effect on the outcome of primary elections, and not simply a reflection of candidate strength, but our conclusions are limited by the quality of the observational data and our assumptions. We assume unconfoundedness after adjusting for the major confounding variables, yet unobserved factors like candidate charisma, campaign strategy, or media exposure could drive both endorsement and victory and bias out estimates upward. Our fundraising variables are limited (only PAC yes/no, not dollar amounts), so more data on fundraising levels and timing, plus media exposure, would strengthen the analysis. Because both the IPW ATE and weighted logistic regression show a statistically significant positive effect, we are fairly confident there is some positive impact of endorsement, but less confident about the exact size. There is uncertainty due to the variability in sampling and model dependence, so this estimated effect should be considered approximate rather than a definitive causal claim. A reasonable alternative explanation for the effect we observed is that Trump is good at backing candidates who were already likely to win.

In terms of prior work, our estimated positive effect of a Trump endorsement on winning is directionally consistent with Boudreau (2016) and Blatte (2024), who also find that endorsements (and Trump's in particular) can meaningfully shift voter support. Unlike those studies, which rely on experiments and surveys, our analysis uses observational data from actual 2022 Republican primaries, so our estimates are noisier and more assumption-dependent but show that similar endorsement effects appear in real election outcomes, using only the causal inference tools covered in this class.

Research Question #2: Multiple Hypothesis Testing/Decision Making

Our second research question via Multiple Hypothesis Testing / Decision Making is as follows: In the 2022 primary elections, are there significant differences in primary win rates across various candidate demographic/positional groups (including gender, race, endorsement, current candidate position, finances, etc)? If so, which differences are statistically significant after multiple comparisons.

Methods:

For this research question, we performed 6 hypothesis tests of differences in primary win rates or shares across groups of candidates political party dependent including: gender (female vs male), race (white vs non-white), incumbency status. The following include our six multiple hypothesis:

- 1) Test 1: Impact of gender on Democratic primary elections → H0: In Democratic primary elections, female and male candidates have the same probability of winning; HA: Their win probabilities are different.
- 2) Test 2: Impact of gender on Republican primary elections → H0: In Republican primary elections, female and male candidates have the same probability of winning; HA: Their win probabilities are different.
- 3) Test 3: Impact of race on Democratic primary elections → H0: In Democratic primary elections, non-white and white candidates have the same probability of winning; HA: Their win probabilities differ.
- 4) Test 4: Impact of race on Republican primary elections → H0: In Republican primary elections, non-white and white candidates have the same probability of winning; HA: Their win probabilities differ.
- 5) Test 5: Impact of incumbency on Republican primary elections → H0: In Republican primary elections, incumbent and non-incumbent candidates have the same probability of winning; HA: Their win probabilities are different.
- 6) Test 6: Impact of incumbency on Democratic primary elections → H0: In Democratic primary elections, incumbent and non-incumbent candidates have the same probability of winning; HA: Their win probabilities are different.

We then use two-sample tests of proportions and logistic regression, depending on whether or not outcomes are binary or continuous. Because we are testing multiple hypotheses in parallel, we apply the Benjamini–Hochberg procedure to control the FDR and minimize the occurrence of Type I errors.

Results:

..	test	type	group1	group0	n1	n0	win_rate1	win_rate0	p_value	bh_q	reject_bh_0.05
0	T1 Dem win-rate Female vs Male	2-proportion z-test	Gender=Female	Gender=Male	379	697	0.509235	0.381636	5.290261e-05	7.406365e-05	True
1	T2 Rep win-rate Female vs Male	2-proportion z-test	Gender=Female	Gender=Male	338	1261	0.272189	0.296590	3.806685e-01	3.806685e-01	False
2	T3 Dem win-rate Non-White vs White	2-proportion z-test	is_white=False	is_white=True	504	572	0.341270	0.501748	1.089463e-07	1.906560e-07	True
3	T4 Rep win-rate Non-White vs White	2-proportion z-test	is_white=False	is_white=True	334	1265	0.224551	0.309091	2.493685e-03	2.909299e-03	True
4	T5 Rep win-rate Incumbent vs Non-Incumbent	2-proportion z-test	is_incumbent=True	is_incumbent=False	216	1383	0.939815	0.190166	1.398446e-112	9.789124e-112	True
5	T6 Dem win-rate Incumbent vs Non-Incumbent	2-proportion z-test	is_incumbent=True	is_incumbent=False	215	861	0.948837	0.296167	4.024356e-67	9.390164e-67	True
6	T6 Dem win-rate Incumbent vs Non-Incumbent	2-proportion z-test	is_incumbent=True	is_incumbent=False	215	861	0.948837	0.296167	4.024356e-67	9.390164e-67	True

We performed a series of two proportion z tests assessing the win rate difference between candidate groups and carried out a Benjamini-Hochberg test tasked with maintaining a false discovery rate of 0.05. Using a false discovery rate of 0.05, 5 out of 6 tests were significant after applying a BH correction.

- Test 1 (Democratic Primary: Female vs. Male win rate)
Female candidates had a higher win rate (0.509, n = 379) than male candidates (0.382, n = 697). This difference was statistically significant ($p = 5.29 \times 10^{-5}$, BH $q = 6.61 \times 10^{-5}$), so we reject the null hypothesis.
- Test 2 (Republican Primary: Female vs. Male win rate)
The win rates for female candidates (0.272, n = 338) and male candidates (0.297, n = 1261) were

very similar. This difference was not statistically significant ($p = 0.381$, BH $q = 0.381$), so we fail to reject the null hypothesis.

- Test 3 (Democratic Primary: Non-White vs. White win rate)
Non-White candidates had a lower win rate (0.341, $n = 504$) compared to White candidates (0.502, $n = 572$). This difference was statistically significant ($p = 1.09\text{e-}07$, BH $q = 1.82\text{e-}07$), so we reject the null hypothesis.
- Test 4 (Republican Primary: Non-White vs. White win rate)
Non-White Republican candidates had a lower win rate (approximately 0.22) than White Republican candidates (approximately 0.31). This difference remained statistically significant after Benjamini–Hochberg correction (BH $q < 0.05$), so we reject the null hypothesis.
- Test 5 (Republican Primary: Incumbent vs. Non-Incumbent win rate)
Incumbent candidates had a much higher win rate (0.939, $n = 216$) than non-incumbent candidates (0.190, $n = 1383$). This difference was highly statistically significant ($p = 1.40\text{e-}112$, BH $q = 6.99\text{e-}112$), leading us to reject the null hypothesis.
- Test 6 (Democratic Primary: Incumbent vs. Non-Incumbent win rate)
Incumbents also had a much higher win rate (0.949, $n = 215$) compared to non-incumbents (0.296, $n = 861$). This result was highly statistically significant ($p = 4.02\text{e-}67$, BH $q = 1.01\text{e-}66$), so we reject the null hypothesis.

Discussion:

On balance, we find by multiple hypothesis testing that some characteristics of candidates are strongly correlated with outcomes in primary elections, while for other characteristics, weaker or irregular relationships with outcomes are found for different parties. Using the Benjamini & Hochberg method to correct for FDR, some differences in win odds were found to be significant.

The gender seems to have implications for the parties differently. In the democratic primaries, the candidates who were female won more than their counterparts who were male. This is an indication that gender could have an association with the outcomes in the democratic elections. In the republican primaries, gender didn't have any significant implications on the outcomes. This is an indication that gender has an association with the outcomes in the republican elections.

Race was a significant predictor of the win rate of candidates in both parties. In both Democratic and Republican primaries, non-White candidates had lower probabilities of winning than White candidates. This point highlights that the disparity in win rates for different races exists regardless of political affiliation.

Incumbency was the strongest and most persistent factor in relation to winning. The chances of winning were considerably higher for incumbent candidates than non-incumbent candidates in Republican primaries, thus asserting the significant advantage of incumbency. This was expected since incumbents enjoy the benefits of greater familiarity and advantage regarding institutional resources.

Although these results are derived from observational studies, one must interpret these findings as correlation, not causation. Furthermore, there may be additional variables, such as candidate quality and district competitiveness, that are also playing a role in these observed margins. In sum, these results identify what kind of demographic and positional characteristics are most strongly associated with primary election victories, correcting for multiple comparisons.

Conclusions

Outcomes Summary:

Across both research questions, we find that structural factors and endorsements matter a great deal for primary outcomes. For Republican candidates, our causal analysis suggests that receiving a Trump endorsement is associated with a higher probability of winning the primary. After reweighting candidates using IPW to adjust for observed confounders, we can see that the endorsement is linked to a positive average treatment effect, and our weighted regression indicates a large, statistically significant increase in win likelihood. For group differences, the multiple hypothesis tests show that incumbency is the strongest and most consistent predictor of winning in both parties. The tests also reveal that non-white candidates have systematically lower win rates than white candidates, and gender relates to outcomes differently by party, with women having an advantage in Democratic primaries but no clear difference from men in Republican primaries.

Critical Evaluation:

Our findings have to be accepted with consideration of several limitations. The data is a census of major primaries and is observational, not random, which means that we cannot apply the same assumptions we would to a fully random sample. The data is also limited in its scope; there are several potentially important variables not being considered, especially in a process as complicated as an election. Our causal interpretation for Trump endorsements depends on an unconfoundedness assumption, but if unobserved factors influence both endorsement and win rate, our estimates may be biased. Though we've accounted for all confounders included in the data, we cannot confirm that there are no other confounders excluded from the data. Similarly, the multiple hypothesis tests are comparisons of win rates which identify associations, but do not completely isolate causal influences. We also lack some domain knowledge that could impact how we navigate the process of formulating and answering these questions; it would be helpful to ask an expert (such as a political scientist or a campaign manager) about what factors they personally believe influences a candidate's chance of winning, and whether or not they think the confounders we take into account are encompassing enough to ensure our results are unbiased. Their answer would either give us more confidence in our results, or give us an idea of how to strengthen our results with supplemental data, consideration for more factors, or other, more rigorous methods. Despite the limitations, our analysis still reveals a lot about election patterns and influences on win rates, and the results can still be fairly and validly interpreted with the acknowledgement of these limitations.

Our conclusions are moderately tied to our modeling choices; one plausible change would be shifting the trimming threshold for propensity scores, which, when changed, could have an impact on the estimated effect size. However, we made these choices with the goal of keeping the data as accurate and unbiased as possible, and other than minor changes in the effect size, the overall trends in the findings should remain the same regardless. In terms of generalizability, our findings are significantly visible, but overall, a bit narrow. Because of the particular dataset we used, our findings apply to major party U.S. primaries from 2022, but not to all general elections, local races, international elections, or past/future election years. We can infer that certain findings, such as the observed relationship between incumbency and win rate, probably generalize to more than just the dataset elections, but other more specific metrics, such as the effect size of a Trump endorsement, is more specifically relevant to the time frame and scope of the dataset we used, and likely more subject to change.

Recommendations

With our results in mind, a natural next step would be to design a follow up study that utilizes a broader dataset, as well as more comprehensive methodologies, further answering the questions our work begins to ask. For example, including data such as fundraising and media coverage, and adjusting for them as possible confounders before estimating the causal effect of a Trump endorsement or the demographic win rate gaps would give the findings more validity. Another angle would be to run similar experiments or surveys that present voters with hypothetical candidates with varying traits, and use voter approval as a metric.

Based on our findings, there are several recommendations to be given to various groups. First, the findings that non white and non incumbent candidates are underperforming is troubling, leading us to encourage voters, donors, and party managers to proactively increase support for said candidates. Our results show consistent racial disparities in win rates for both parties, revealing a need to level the playing field for non-white candidates who are as qualified as their white counterparts. As a concrete call to action, this could include encouraging training and awareness for political party managers and donors, increased fundraising efforts for minority candidates, and an emphasis on policy over demographic data pushed in political advertisements and debates. From an ethical standpoint, acknowledging systematic gaps is crucial to avoiding the othering and mistreatment of minority groups. In a political setting, equality and fairness is especially crucial, as the outcomes of elections influence millions of people.

In relation to the results of our study, future studies that could build up on our work include an analysis of the immediate effects of Trump endorsements by accounting for timing of the exact dates which they are given. This study would then involve comparing outcomes for candidates in terms of shares, fundraising, polls immediately before and after the endorsement. This would control for the features of the candidate itself (example strengths, skills) and measure changes after endorsement more directly. Future studies could also expand the dataset to more election years to see if findings from 2022 are reflective of a generalized approach to findings from 2022.

Works Cited

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